

Test Case: WB-01

Path: P1

Input:

```
{
  "request_id": "REQ-01",
  "blood_group": "A+",
  "units": 5,
  "hospital_id": "H-001"
}
```

Expected Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated."
}
```

[Notification] Hospital H-001 notified. Request REQ-01 status: Allocated

Actual Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated.",
  "allocatedUnits": 5,
  "bloodGroup": "A+",
  "hospitalId": "H-001"
}
```

Result: PASS

Test Case: WB-02

Path: P2

Input:

```
{
  "request_id": "REQ-02",
  "blood_group": "B+",
  "units": 1,
  "hospital_id": "H-001"
}
```

Expected Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated."
}
```

[Alert] SHORTAGE ALERT: Low stock for blood group B+!

[Notification] Hospital H-001 notified. Request REQ-02 status: Allocated

[Notification] Hospital H-001 notified. Request REQ-02 status: Allocated

Actual Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated.",
  "allocatedUnits": 1,
  "bloodGroup": "B+",
  "hospitalId": "H-001",
  "alert": "shortage alert triggered"
}
```

Result: PASS

Test Case: WB-03

Path: P3

Input:

```
{
  "request_id": "REQ-03",
  "blood_group": "",
  "units": 5,
  "hospital_id": "H-001"
}
```

Expected Output:

```
{
  "status": "ERROR",
  "message": "Blood group required"
}
```

Actual Output:

```
{
  "status": "ERROR",
  "message": "Blood group required",
  "allocatedUnits": 0,
  "bloodGroup": "",
  "hospitalId": "H-001"
}
```

Result: PASS

Test Case: WB-04

Path: P3

Input:

```
{
  "request_id": "REQ-04",
  "blood_group": "INVALID",
  "units": 5,
  "hospital_id": "H-001"
}
```

Expected Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated."
}
[Alert] SHORTAGE ALERT: Low stock for blood group B+!
[Notification] Hospital H-001 notified. Request REQ-15 status: Allocated
```

Actual Output:

```
{
  "status": "SUCCESS",
  "message": "Blood units successfully allocated.",
  "allocatedUnits": 5,
  "bloodGroup": "B+",
  "hospitalId": "H-001",
  "alert": "shortage alert triggered"
}
```

Result: PASS

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WHITE BOX TEST SUMMARY

Total Tests: 15

Passed: 15

Failed: 0

Coverage:

- Statement Coverage: 100%
- Branch Coverage: 100%
- Path Coverage: P1, P2, P3 covered

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```
exam3@sysadmin-HP-ProOne-440-23-8-inch-69-All-in-One-Desktop-PC:~/Desktop/56 lab exam$ 5-[BVA: Units = 11] Result:[BVA: Units = 11]
# Outputs:
```